2	**(a)**	$\Delta E = mg\Delta h$	C1	
		$= 0.030 \times 9.81 \times (–)0.31$		
		$= (–)0.091$ J	A1	[2]
	(b)	$E = ½mv^2$	C1	
		(initial) $E = ½ \times 0.030 \times 1.3^2$ (= 0.0254)	C1	
		$0.5 \times 0.030 \times v^2 = (0.5 \times 0.030 \times 1.3^2) + (0.030 \times 9.81 \times 0.31)$ so $v = 2.8\,\mathrm{m\,s^{-1}}$ *or* $0.5 \times 0.030 \times v^2 = (0.0254) + (0.091)$ so $v = 2.8\,\mathrm{m\,s^{-1}}$	A1	[3]
	(c)	**(i)** $0.096 = 0.030\,(v + 2.8)$	C1	
		$v = 0.40\,\mathrm{m\,s^{-1}}$	A1	[2]
		(ii) $F = \Delta p / (\Delta)t$ *or* $F = ma$ $= 0.096 / 20 \times 10^{-3}$ *or* $0.030\,(0.40 + 2.8) / 20 \times 10^{-3}$	C1	
		$= 4.8$ N	A1	[2]
	(d)	<u>kinetic</u> energy (of ball and wall) decreases/changes/not conserved, so inelastic *or* (relative) speed of approach (of ball and wall) not equal to/greater than (relative) speed of separation, so inelastic.	B1	[1]
	(e)	force = work done / distance moved $= (0.091 – 0.076) / 0.60$	C1	
		$= 0.025$ N	A1	[2]